BANK DEPLOYMENT REFERENCE

YP-Raksha On-Premises Deployment Architecture

Reference design for 120,000 endpoints across DC, DR and UAT

Document information

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Audience Enterprise architecture, security and operations teams

Release status Validate against the selected product release before procurement or production use

Product capability and sizing statements are labelled by evidence and assumption. Confirm release-specific support and commercial terms with the product owner.

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Scope: primary Data Centre (DC), Disaster Recovery site (DR), and User Acceptance Testing (UAT)

Planning population: 120,000 managed endpoints

Status: reference design and initial sizing proposal, not a measured performance certification

Design objective

Deploy the Gateway and Browser AI Guard inside bank-controlled infrastructure in India. Keep prompts, responses, Browser AI events, application logs, audit records, metrics, traces, backups, and support bundles within the bank's India boundary.

This outcome depends on both configuration and network enforcement. The product can connect to public model providers and optional external observability, email, analytics, or package services. Those options must be disabled or pointed at internal services. If employees can reach public AI websites directly, their prompts may leave the bank even if Guard is installed. A strict no-egress design blocks those destinations and offers approved AI access through the bank-hosted Gateway and private model services.

Data flow and sovereignty boundary

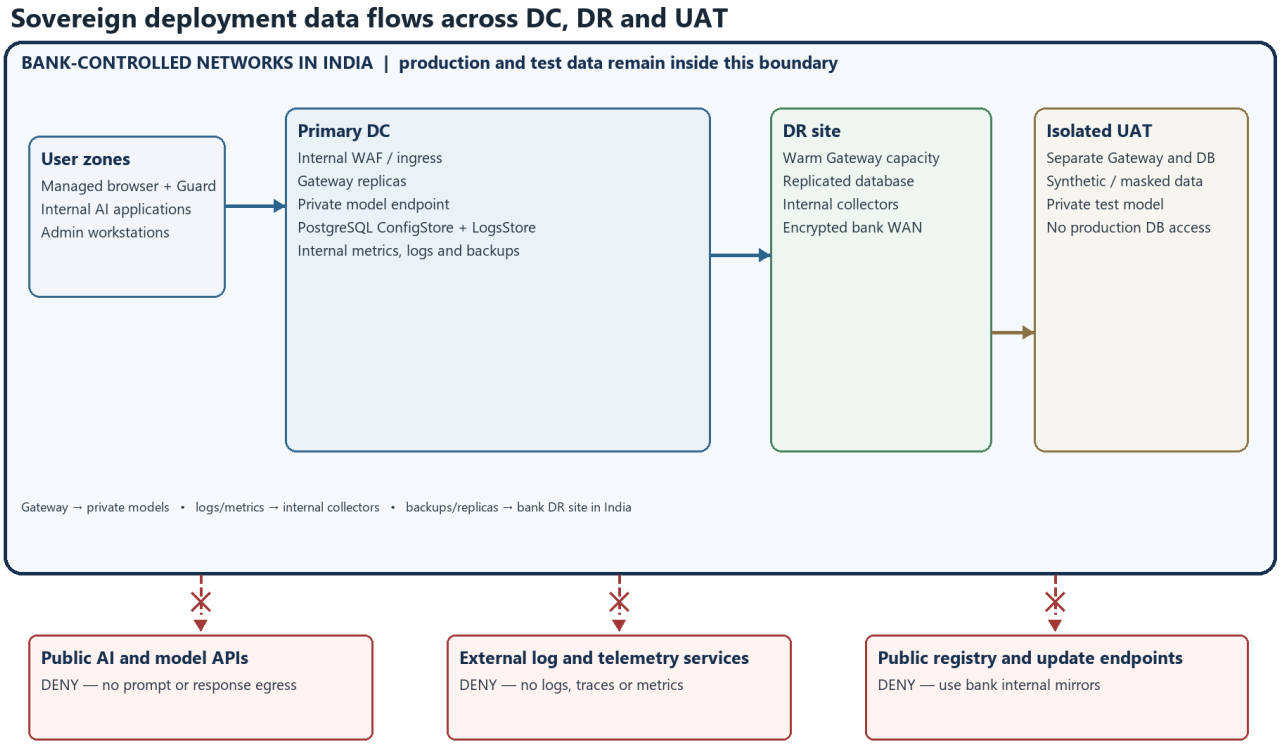


Figure 1. Intended flows remain inside the bank-controlled India boundary; red dashed arrows are denied at the network edge.

The boundary includes the bank-operated DC, DR, endpoint, and UAT networks located in India. DC-to-DR replication remains within that boundary over an encrypted bank WAN. UAT uses synthetic or masked data and has no production database access. Public AI and external telemetry paths are blocked.

Capacity assumptions for 120 000 endpoints

Use these values to plan a first infrastructure baseline:

Assumption	Planning value
Managed Guard endpoints	120,000
Busy-period active endpoints	10% = 12,000
Browser Guard event rate	One reported event per active endpoint every 180 seconds
Calculated steady Guard event rate	12,000 ÷ 180 = about 67 events/second
Planned Guard ingestion rate	150 events/second, allowing roughly 2× burst headroom
AI inference rate assumption	One request per active user every 120 seconds = 100 requests/second
Planned inference rate	200 requests/second, allowing roughly 2× burst headroom

These rates are separate workload classes. Long-running model generations, streaming, large uploads, payload logging, MCP calls, and model compute can change the required capacity substantially. This sizing covers the Gateway/control and storage tiers, not the GPU/CPU capacity of the private model service. Validate with representative prompts, response sizes, session counts, and failure scenarios.

Initial infrastructure sizing

Sizing is for one full-capacity production site plus a warm DR site capable of scaling to the full planning load. CPU and memory are per VM/node. Kubernetes worker counts are capacity pools, not fixed product requirements.

Tier	DC baseline	DR baseline	UAT baseline
Ingress/WAF	2 × 8 vCPU / 16 GiB, active-active	2 × 8 vCPU / 16 GiB, standby/active on failover	2 × 4 vCPU / 8 GiB
Gateway worker pool	4 × 32 vCPU / 128 GiB worker nodes; start with 6 replicas at 8 vCPU / 16 GiB each; autoscale to 12 replicas	4 × 32 vCPU / 128 GiB nodes reserved; start with 3 replicas and scale to 6 on failover	3 × 16 vCPU / 64 GiB nodes; 3 replicas at 4 vCPU / 8 GiB
PostgreSQL HA	3 × 32 vCPU / 256 GiB; primary plus 2 replicas; 4 TiB usable database storage	3 × 32 vCPU / 256 GiB; replicated full dataset; 4 TiB usable	3 × 16 vCPU / 64 GiB; 1 TiB usable
Redis/cache HA, if enabled	3 × 8 vCPU / 32 GiB	3 × 8 vCPU / 32 GiB	3 × 4 vCPU / 16 GiB
Internal monitoring/log collection	3 × 8 vCPU / 32 GiB for bank-operated metrics, tracing, and log collection	3 × 8 vCPU / 32 GiB or equivalent DR capacity	2 × 4 vCPU / 16 GiB
Backup/object storage	At least 8 TiB usable, encrypted and bank-local	At least 8 TiB usable, encrypted and bank-local	At least 2 TiB usable

Use the Gateway's Kubernetes resource requests/limits and HPA to allocate the stated replica envelope. The chart's bundled database/cache examples are useful for evaluation; final HA and cross-site replication should use an operator-approved topology. Keep DC and DR capacity reservations available if the failover objective requires full 120,000-endpoint load at the DR site.

Storage calculation

For a 90-day retention illustration, assume five Guard/log events per endpoint per day and a 4 KB average normalized event:

- 120,000 × 5 = 600,000 events/day.
- 600,000 × 4 KB = 2.4 GB/day raw.

- 90 days = 216 GB raw.
- Reserving 2.5× for indexes, write-ahead logs, and working overhead yields about 540 GB before growth reserve.

The 4 TiB database allocation provides room for growth and other Gateway records under this narrow event-size assumption. Full prompt/response payloads, uploaded files, higher event rates, longer retention, and replication/backup copies increase storage requirements; measure actual event sizes and recalculate before procurement. Replicas and backups require their own site capacity.

Site roles and recovery

DC

- Serve production management, inference, and Guard event APIs behind redundant internal ingress.
- Run Gateway replicas across failure domains and use bank-operated HA database/cache services.
- Keep model endpoints, DNS, identity, SMTP relay, image/package mirror, metrics, logs, and backup targets internal.
- Enforce application and network policies so database and operations services are not reachable from user networks.

DR

- Maintain replicated configuration and log data, private model dependencies, internal DNS, and tested Gateway images.
- Keep sufficient reserved capacity to scale Gateway replicas from the warm level to the planned production level.
- Candidate planning objectives are RPO ≤ 15 minutes and RTO ≤ 60 minutes. These are targets for engineering and exercise, not product guarantees. If the bank requires RPO 0, select and test synchronous replication with the expected WAN latency and performance impact.
- Document database promotion, DNS/route cutover, key/secrets availability, and failback. No automatic cross-site DR guarantee is implied by the Gateway cluster feature.

UAT

- Isolate UAT from production databases and production secrets.
- Use synthetic or masked records, separate credentials, a private test model endpoint, and internal-only telemetry.
- Exercise the selected endpoint OS, browser versions, VDI image, PAC/certificate distribution, policy behavior, peak workload, retention, and DC-to-DR recovery procedure before production rollout.

Endpoint browser and VDI support planning

Platform	Status for this design
Windows endpoints	Browser Guard packaging is present for Windows. Confirm exact supported Windows editions/builds against the release installer in UAT.
macOS endpoints	Browser Guard packaging is present for macOS. Confirm exact versions, permissions, signing/notarization, and certificate deployment in UAT.
Linux endpoints	No Linux Guard endpoint installer is documented. Linux is used for Gateway container/Kubernetes infrastructure.
Chrome and Edge	Candidate browsers for PAC/system-proxy deployment; test selected versions and enterprise certificate policy.
Firefox	Candidate only; validate PAC behavior and Firefox-specific certificate trust distribution.
Safari	Candidate only on macOS; validate system proxy and certificate trust behavior.

Platform	Status for this design
Citrix, VMware Horizon, Azure Virtual Desktop, other VDI	No named VDI platform is certified in the repository. Validate persistent/non-persistent image behavior, per-session identity, PAC settings, CA trust, and agent lifecycle in UAT before calling it supported.

The product repository does not provide a versioned browser or VDI certification matrix. Record the tested OS/browser/VDI versions in the bank's release acceptance record.

Network zones and permitted flows

Keep these flows internal to the bank's India network:

Source → destination	Purpose	Typical protocol/port
Endpoints/VDI → internal ingress	Gateway UI/API and Guard event/config traffic	HTTPS 443
Ingress → Gateway replicas	API and WebSocket/realtime traffic	Internal HTTPS or approved service port
Gateway → PostgreSQL	Configuration and log persistence	TCP 5432 or bank-selected port
Gateway → Redis/cache	Cache or distributed state, if enabled	TCP 6379 or bank-selected port
Gateway → private model service	Inference	Internal HTTPS
Gateway → internal telemetry/metrics	Metrics and traces	Internal scrape/OTLP ports as configured
DC → DR	Encrypted database and backup replication	Bank-approved private WAN and database ports
Admin devices → management ingress	Administration	HTTPS 443

Ports are examples and must match the selected deployment and firewall policy. Browser Guard's local proxy/PAC listener is endpoint-local; the optional shared proxy is a separate internal service and needs its own capacity validation.

Zero egress configuration and evidence

Apply all of these controls together:

1. **Default-deny egress:** block public Internet egress from Gateway, Guard proxy, database, and telemetry networks. Permit only named internal destinations. Deny public DNS, public AI-provider APIs, public analytics, and direct SMTP.
2. **Private AI only:** configure provider adapters only for bank-hosted or bank-approved private endpoints. Block public AI website domains at the bank proxy/firewall. Do not treat prompt redaction alone as proof that prompts remain in-country.
3. **Internal telemetry only:** disable external exporters/connectors or target bank-hosted collectors. Restrict the metrics endpoint to internal scrape sources. Check webhook, alert, log-shipping, and analytics destinations.
4. **Offline/internal dependencies:** mirror container images, Helm charts, binaries, model parameter catalogs, and updates inside the bank. Set the Helm image repository to the bank registry and override the example public model-parameters URL with an internal file or URL. The repository's Compose example specifies public DNS resolvers; replace those with bank DNS before deployment.
5. **Internal operations:** use bank DNS, NTP, identity/SCIM, SMTP relay, secrets management, and backup/object storage. Confirm the commercial license/activation path supports the same offline boundary.
6. **Protect records:** define payload logging and retention, encrypt database and backup storage, restrict access, and prevent support bundles from being copied outside the bank.

7. **Verify at the network boundary:** review firewall/NAT deny and allow logs, DNS query logs, Kubernetes egress policy, and packet captures during startup, normal use, updates, inference, telemetry, and failure recovery. Demonstrate that attempted external destinations fail while approved internal flows work.

The diagram represents the intended flow under these enforced controls. Application configuration alone cannot prove that no traffic leaves; the bank's firewall, DNS, proxy, and operational evidence provide that proof. Do not approve the no-egress claim until external paths have been tested and denied.

Production acceptance checklist

- Confirm 120,000 endpoints are registered and 150 Guard events/second plus 200 inference requests/second are sustained under representative payloads.
- Confirm database write latency, connection pools, retention cleanup, backup/restore, and storage growth.
- Confirm DC node, database, cache, ingress, and WAN failures do not create unacceptable data loss or downtime.
- Execute the DR promotion and failback runbook and record observed RPO/RTO.
- Confirm endpoint updates and VDI image lifecycle do not disable proxy policy or duplicate device identities.
- Verify every model, telemetry, log, alert, email, identity, and update destination remains inside the approved boundary.
- Obtain sign-off from network security, platform operations, database operations, endpoint engineering, and data-residency owners.

Repository references

- Technical Documentation
- Product Datasheet
- Server and deployment guide
- Helm chart
- Docker Compose example
- Browser Guard packaging guide

Detailed service topology and responsibilities

Zone or service	Runs in	Primary responsibility	Owner in this design
Endpoint and VDI zone	Bank-managed user networks in India	Browser, local Guard agent, enterprise certificate and PAC configuration	Endpoint engineering and workplace operations
Application zone	Bank data centre network in India	Internal AI applications and Gateway clients	Application owners
Ingress zone	DC, DR and UAT internal networks	TLS termination, WAF/load balancing, health checks and restricted routing	Network/security operations
Gateway application zone	Kubernetes or container platform	API, policy, provider routing, admin UI, Browser AI control plane and metrics endpoint	Platform operations
Data zone	Bank database network	ConfigStore, LogsStore, cache if enabled, database backups	DBA and storage operations
Private model zone	Bank or approved private service network	Inference execution and model artifacts	AI platform/model operations
Operations zone	Bank internal management networks	Identity, secrets, DNS, NTP, log/metrics/tracing, package mirror and alert relay	Shared infrastructure operations

Zone or service	Runs in	Primary responsibility	Owner in this design
DR site	Separate bank site in India	Replicated data, reserved application capacity and internal dependencies	Platform, DBA, network operations
UAT zone	Separate bank-managed environment	Release qualification with synthetic or masked data	QA/UAT and security owners

Use explicit network policies and firewall rules between zones. A broad network route inside the bank is not a substitute for a least-privilege flow policy.

Data classification and storage paths

Data class	Examples	Expected path in the design	Residency control
AI prompt and response	Application inference input/output, browser-submitted content when captured	Client → internal ingress → Gateway → private model; configured records → LogsStore	Private model endpoint, internal database, payload logging decision and firewall deny for external model APIs
Identity and credentials	User, virtual key, provider and service credentials	Admin/identity integration → Gateway/ConfigStore and bank secrets service	Restricted admin roles, encryption/key management, no secrets in UAT production copies
Browser AI event	Agent identity, target, timestamps, verdict and any configured captured content	Endpoint/shared proxy → internal Gateway → Browser AI store	Internal ingress only, content minimization, retention policy, internal backups
Audit and operational logs	Admin actions, service logs and security events	Gateway/cluster → internal log collector and LogsStore	Internal collectors, access controls and egress deny
Metrics and traces	Health, counters, timings and configured trace attributes	Gateway → internal Prometheus/OTLP collector	Internal scrape/export targets; disable public telemetry integrations
Backups and replicas	Database and object-store recovery data	DC database/storage → encrypted bank WAN → DR storage	Both sites located in India, encryption, access control and restore testing
Build and update artifacts	Containers, Helm charts, Guard installers, model catalog files	Approved internal mirror/repository → DC/DR/UAT	Scan and promote from controlled import; no runtime pull from public registry

Retention duration is a bank policy decision. Set deletion and archive schedules for each class, including backups, database replicas, object storage and exported operational records. The product's ability to log content makes data minimization a configuration decision as well as a legal and operational requirement.

Reference flow sequence

1. A managed application or Guard-enabled browser sends traffic to the internal ingress over TLS. Browser destinations not approved for Guard are either blocked or handled according to bank policy; users must not retain a direct public path around the Gateway.
2. The ingress routes approved requests to a healthy Gateway replica. The Gateway authenticates the request, applies configured authorization, policy and usage controls, and selects an approved destination.
3. An inference request goes to a private model endpoint in a bank-controlled or approved India network. No external provider API is configured in the strict no-egress profile.
4. The Gateway writes configured operational and inference records to internal persistence. The browser agent sends control-plane/event traffic to the internal Gateway; event details are retained according to policy.

5. Metrics and logs are collected by bank-operated internal collectors. DC database and backup data replicate to the bank's DR site within India using encrypted site-to-site connectivity.
6. Egress firewalls deny public model, telemetry, logging, analytics, email and package destinations. The bank verifies the deny with DNS, firewall and packet-capture evidence during representative operations.

Detailed capacity model

Workload assumptions

Workload dimension	Reference assumption	Capacity implication
Registered endpoints	120,000	Fleet inventory and periodic heartbeat should be tested independently from user traffic
Busy-period active fraction	10% (12,000 endpoints/users)	Initial planning factor only; use measured concurrency if available
Reported Guard events	One event per active endpoint every 180 seconds	Approximately 67 events/second steady-state
Guard ingestion target	150 events/second	Roughly 2.2× steady-state planning headroom; verify with representative event size
AI inference requests	One per active user every 120 seconds	Approximately 100 requests/second steady-state
Inference planning target	200 requests/second	2× steady-state planning headroom; streaming concurrency must be modelled separately
Retention illustration	5 events/endpoint/day, 4 KB/event, 90 days	Approximately 216 GB raw, approximately 540 GB at 2.5× for indexes/WAL/working overhead
Database capacity reservation	4 TiB usable per production site	Planning allocation; not a claim that all workloads fit without measurement

The retention illustration is deliberately narrow: it excludes large prompt/response bodies, files, attachments, backups, replicas, model data and long-lived traces. Measure actual bytes per stored event and keep site replicas/backups in their own storage calculations.

Resource envelope and interpretation

The listed worker capacity is a starting reserve for load, rolling maintenance and node loss. Pod requests are scheduled on those worker pools; they are not additional hardware to add on top of the worker-node totals. For each site, leave capacity for the platform system pods, OS, monitoring agents and failover headroom. The database and model service pools are separate from the Kubernetes application workers unless the bank explicitly chooses a shared platform.

At full reservation, the DC and DR each include 128 vCPU / 512 GiB of Gateway worker nodes, plus separate ingress, database, cache (if enabled) and observability hosts from the sizing table. Gateway pods start at 6 × 8 vCPU / 16 GiB at DC and 3 × 8 vCPU / 16 GiB at warm DR; the intended maximum is 12 replicas at DC or 6 at DR failover. Kubernetes scheduling, HPA behavior and provider connections must be proven under the bank's topology.

UAT has 3 × 16 vCPU / 64 GiB Gateway workers (48 vCPU / 192 GiB) and starts with 3 × 4 vCPU / 8 GiB Gateway replicas. This supports release qualification and functional/performance rehearsal, not production-equivalent throughput unless separately scaled.

These estimates exclude GPU/CPU requirements for private model inference, endpoint fleet-management servers, enterprise network/WAF shared capacity, directory services, internal artifact registries, and any separate high-volume log platform. Obtain model-vendor sizing from the actual model, context window, token rate, concurrency and accelerator plan.

Database cache backup and retention design

- Use a bank-approved PostgreSQL HA topology for the reference deployment. The application repository includes database drivers, but cross-site replication, database failover and storage durability are infrastructure responsibilities.
- Place database nodes in separate failure domains where the bank platform supports them. Configure TLS, authentication, connection limits, monitoring, backups, WAL archiving/replication and restore procedures under DBA standards.
- Redis/cache is optional. If enabled, use a bank-operated HA design and define whether loss of cache affects availability, latency, rate limits or distributed state. Do not treat it as the source of truth for durable configuration or logs unless explicitly supported by the release design.
- Define table retention, cleanup, archival and deletion schedules. Verify scheduled jobs and database growth after enabling payload or Browser AI content capture.
- Encrypt backup media and copies, keep them in India, restrict restore operators and test point-in-time recovery. Apply separate retention to immutable backups and replicas.
- Back up the secrets needed to restore service, including database credentials, encryption keys, Guard shared secrets, TLS materials and provider/model credentials, using the bank's segregated secrets process. A database backup without compatible key recovery may not be usable.

Environment lifecycle and change control

Promote an immutable release artifact from internal registry through UAT to DR and DC. Record the artifact digest, configuration version, schema migration, package scan result, approval, rollback version and change ticket. Do not have production nodes reach public registries or update endpoints at runtime.

For a disconnected update process, import signed/approved artifacts through the bank's controlled transfer process, scan them, store them in the internal registry and test them in UAT. The product repository contains defaults that require environment overrides: the Helm example points to a public image registry and a public model-parameter catalog URL; the Compose example uses public DNS resolvers; sample SMTP settings use an external mail service. Replace these before deployment. Confirm offline licence activation and Guard-update behavior with the product owner before accepting a fully disconnected environment.

Disaster recovery operating model

The DC/DR objective is a bank-operated recovery pattern, not automatic product-level site failover. The following sequence is a reference runbook outline:

1. Declare site incident and freeze configuration changes according to the bank incident process.
2. Confirm DR database recovery point, replica health, encryption-key availability and restore integrity.
3. Promote the DR database using the DBA-approved mechanism; validate application schema and service accounts.
4. Scale Gateway replicas to the approved failover count; confirm ingress, TLS certificates, secrets, private model reachability, DNS, identity and internal collectors at DR.
5. Move approved internal DNS/load-balancer routes to DR and monitor health, error rates, database saturation, queue/backlog and model capacity.
6. Reconcile endpoint connectivity and agent heartbeat state; handle duplicate/replayed event records according to the product's idempotency behavior established during testing.
7. Record observed RPO/RTO. Do not claim the candidate target until an exercise has demonstrated it.
8. Plan failback with data reconciliation, route cutback, replica re-seeding and a second approved change window.

Candidate objectives of RPO ≤15 minutes and RTO ≤60 minutes require validation. Zero-data-loss recovery generally requires synchronous replication and testing of WAN latency, commit latency and site failure behavior; it must not be inferred from this reference size.

Rollout and capacity qualification

A staged rollout reduces the chance of discovering endpoint policy or identity collisions at fleet scale. A suggested bank rollout sequence is: representative IT pilot, controlled business pilot, 5,000 endpoints, 20,000, 60,000 and then the full 120,000. These are program checkpoints, not product-mandated batch sizes. Proceed only after security, endpoint, application and operations owners accept the prior stage.

For each stage, measure endpoint heartbeat distribution, event burst rate, request sizes, TLS/session counts, inference rate, response streaming concurrency, DB connections, write latency, storage growth, error rate and proxy bypass behavior. Include cold-start, restart, rolling update, certificate renewal, collector outage, DB failover and DC-to-DR test cases. Load-test with synthetic data and representative payloads; do not generate production prompts in UAT.

Monitoring and alerting baseline

Collect at least: Gateway health and restart counts; request rate, latency and errors by route/provider/model; active and failed streams; database pool saturation, write latency and replication lag; storage consumption and growth; Guard agent heartbeat and version distribution; Guard event rate and failed event delivery; ingress saturation; egress denies; backup completion and restore status; and DR replication/failover health.

Metrics labels and trace attributes can contain sensitive identifiers. Inspect and minimize labels, restrict `/metrics` and tracing endpoints to internal collectors, and verify that alert channels point to internal destinations. Public or third-party dashboards require a separate data residency approval and are excluded from the strict no-egress design.

Network policy matrix

Source	Destination	Allow condition	Example port/protocol	Data/control carried
Managed endpoints/VDI	Internal ingress VIP	Approved clients and Guard configuration only	TCP 443 HTTPS	Requests, Guard heartbeat/config/events, administration where authorized
Internal applications	Internal ingress VIP	Application identity and approved virtual key	TCP 443 HTTPS	Inference requests and responses
Admin workstations	Management ingress	Privileged access segment and MFA/identity policy	TCP 443 HTTPS	Configuration and administrative records
Ingress/WAF	Gateway service	Healthy service endpoints only	Bank-selected HTTPS/service port	Routed API traffic
Gateway pods	PostgreSQL service	Approved service accounts and TLS	TCP 5432 or bank-selected	Configuration, logs, audit and Browser AI data
Gateway pods	Redis/cache	Only if enabled	TCP 6379 or bank-selected	Cache/distributed state
Gateway pods	Private model endpoint	Model allow-list; bank internal route	Internal HTTPS	Prompt/input and model response
Gateway pods	Internal collectors	Only configured metrics/log/trace paths	Internal scrape/OTLP ports	Operational metadata and metrics
DC database/backup	DR database/backup	DBA-approved encrypted replication/backup flow	Bank-selected private WAN ports	Replication and backup data

Source	Destination	Allow condition	Example port/protocol	Data/control carried
Gateway/agent subnets	Public Internet	Deny by default; exceptions require explicit residency approval	Deny	No public model, telemetry, analytics, log export or direct package pull

Ports and addresses are examples only. The deployment owner must publish the actual source/destination CIDRs, protocol, identity, direction, owner and expiry for each exception in the bank firewall request.

No egress enforcement and proof

The diagram and flow tables describe the intended boundary. Application configuration alone cannot prove that data remains in India. The bank's technical enforcement and evidence must include:

1. Default-deny outbound policy from endpoint, Gateway, database, observability and UAT zones, with an allow-list limited to approved bank-owned internal services.
2. Internal private model endpoints; no public provider credential, external model URL or unapproved internet route in production configuration.
3. Internal DNS resolvers and monitoring for DNS attempts to public AI, telemetry, analytics, SMTP, licensing and update domains.
4. Internal Prometheus/OTLP/log collectors; disable or redirect third-party tracing, analytics, webhook, email, object-store and log-export integrations.
5. Internal container/Helm/binary/Guard/model-catalog mirrors. Override public image and model catalog URLs in chart defaults; replace Compose public DNS and sample external SMTP settings.
6. Browser proxy/firewall controls that prevent users from bypassing Guard and directly reaching unapproved public AI websites. Endpoint configuration by itself does not constrain unmanaged routes.
7. Firewall deny logs, DNS logs, Kubernetes egress-policy review and packet capture covering startup, login, inference, event reporting, updates, alert delivery, backup, failover and failure cases.
8. A data-flow inventory signed by the product owner, bank network security, data-residency owner and application/model owners, including any license activation or support procedure.

A site that permits any outside destination cannot make an unconditional statement that no traffic leaves. Record allowed egress exceptions explicitly, identify data classes that could flow through them, and obtain the bank's approval before production.

UAT and production acceptance criteria

- Demonstrate production and DR are physically or logically located within India and show the applicable network and hosting boundaries.
- Confirm endpoint OS/browser/VDI versions and exact Guard package builds in the bank's acceptance record.
- Sustain the planning load of 150 Guard events/second and 200 inference requests/second with representative request sizes, streaming mix and internal model latency; publish test conditions and results.
- Verify database failover, pool sizing, replication lag, backups, restore and retention deletion.
- Exercise a DC-to-DR cutover and failback; report observed RPO and RTO against bank-approved targets.
- Confirm users cannot directly reach unapproved public AI services and that approved internal model routing succeeds.
- Verify agent identity persistence in persistent and non-persistent endpoints/VDI, proxy policy enforcement, certificate trust renewal and update rollback.
- Confirm all log, trace, metric, alert, email, model, license, package and support destinations are internal or disabled.
- Capture firewall, DNS, route, packet-capture, database, application and operational evidence proving the approved flow set.

- Obtain sign-off from platform, endpoint, network security, DBA, application, model, operations, data-residency and compliance owners.

Open decisions for the bank and product owner

Decision	Required before
Exact product release, artifact digest and supported-version matrix	UAT build freeze
Browser/OS/VDI qualification list	Endpoint pilot
Provider/model API operations and concurrency profile	Application onboarding and load test
Payload and event content logging policy	Production data approval
Database version/topology and cross-site replication mode	Procurement and DR test
RPO/RTO and synchronous/asynchronous recovery choice	DR design approval
Offline license activation and update process	No-egress sign-off
Public egress exceptions, if any	Production security approval
Commercial support, service hours, SLA and certification claims	Contract signature

References

- Technical Documentation
- Product Datasheet
- Server and Deployment Documentation
- Administrator Guide
- Helm values
- Docker Compose example
- Browser Guard packaging guide